Power input (USB–C) and regulators

File: lcd_dual_power.kicad_sch

HDMI receiver LT8619C

File: lcd_dual_hdmi.kicad_sch

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd_dual_panel.kicad_sch

STM32F072 controller and GT915 touch

File: lcd_dual_mcu.kicad_sch

AT070TN92 7'' 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch –> USB HID (STM32F072).

Power: USB–C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

U10: STM32F072CBT6 or CH32X035C8T6 (same LQFP–48 footprint). SW2 = CH32 USB bootloader (D+ high at power–up).

Panel power–up order (MCU controlled): DVDD –> RESET –> VGL/AVDD –> VGH –> data –> backlight.

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP–contact 0.5 mm 50–pin FPC connector.

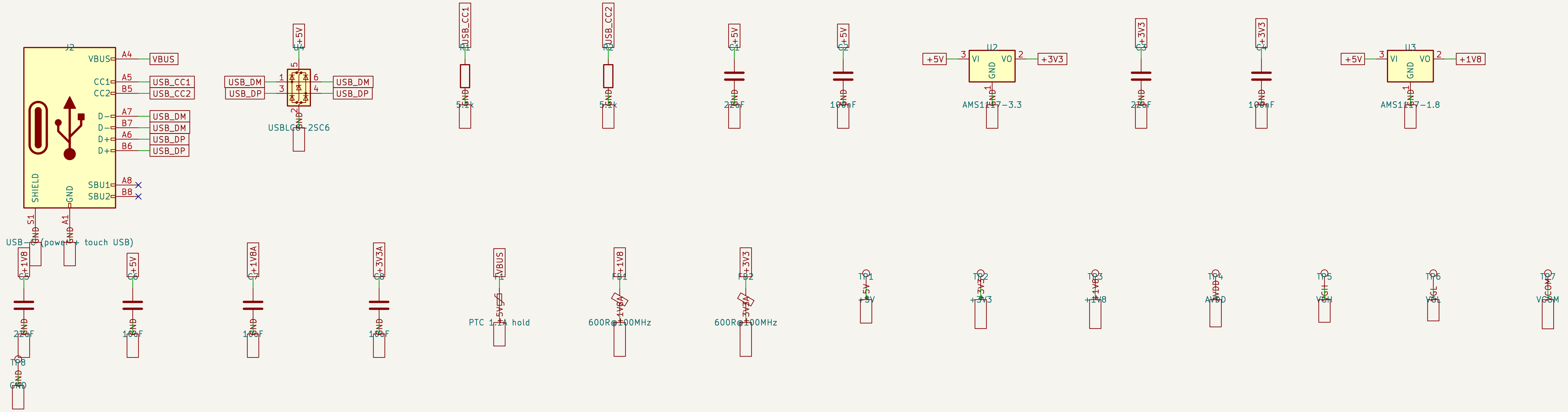
VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

LT8619C HDMI–>RGB888, TPS61040 bias, TPS61165 backlight, STM32F072 USB touch

Sheet: /
File: lcd_dual.kicad_sch

Title: AT070TN92 HDMI + USB–touch driver board

Size: A4	Date: 2026–10–02	**Rev: 1.0**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/5



Generated from hardware/gen/design.py — edit the generator, not this file

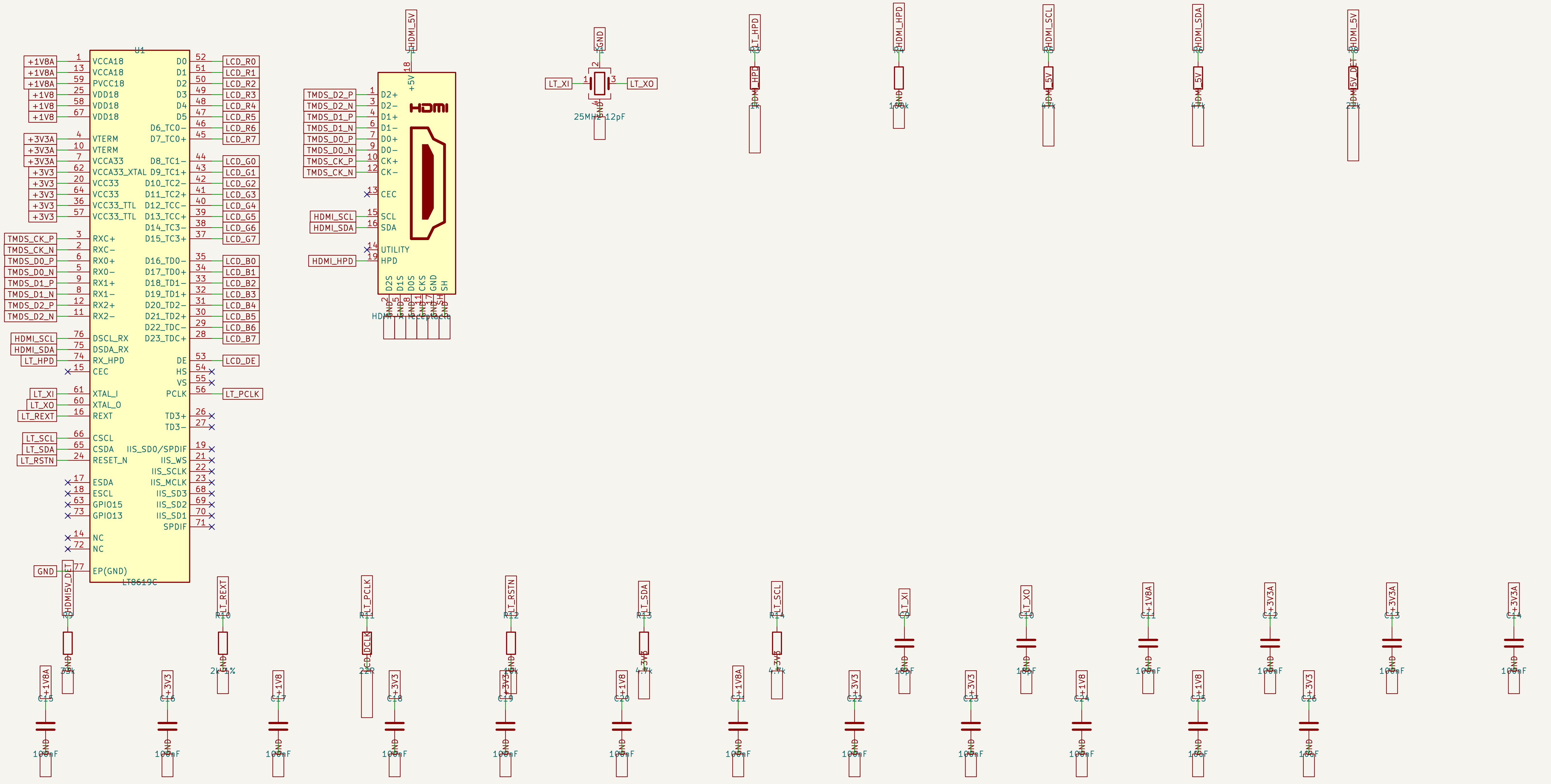
AT070TN92 HDMI + USB touch driver

Sheet: /Power input (USB−C) and regulators/
File: lcd_dual_power.kicad_sch

Title: Power input (USB−C) and regulators

Size: A3 Date: 2026−10−02 **Rev: 1.0**

KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 2/5



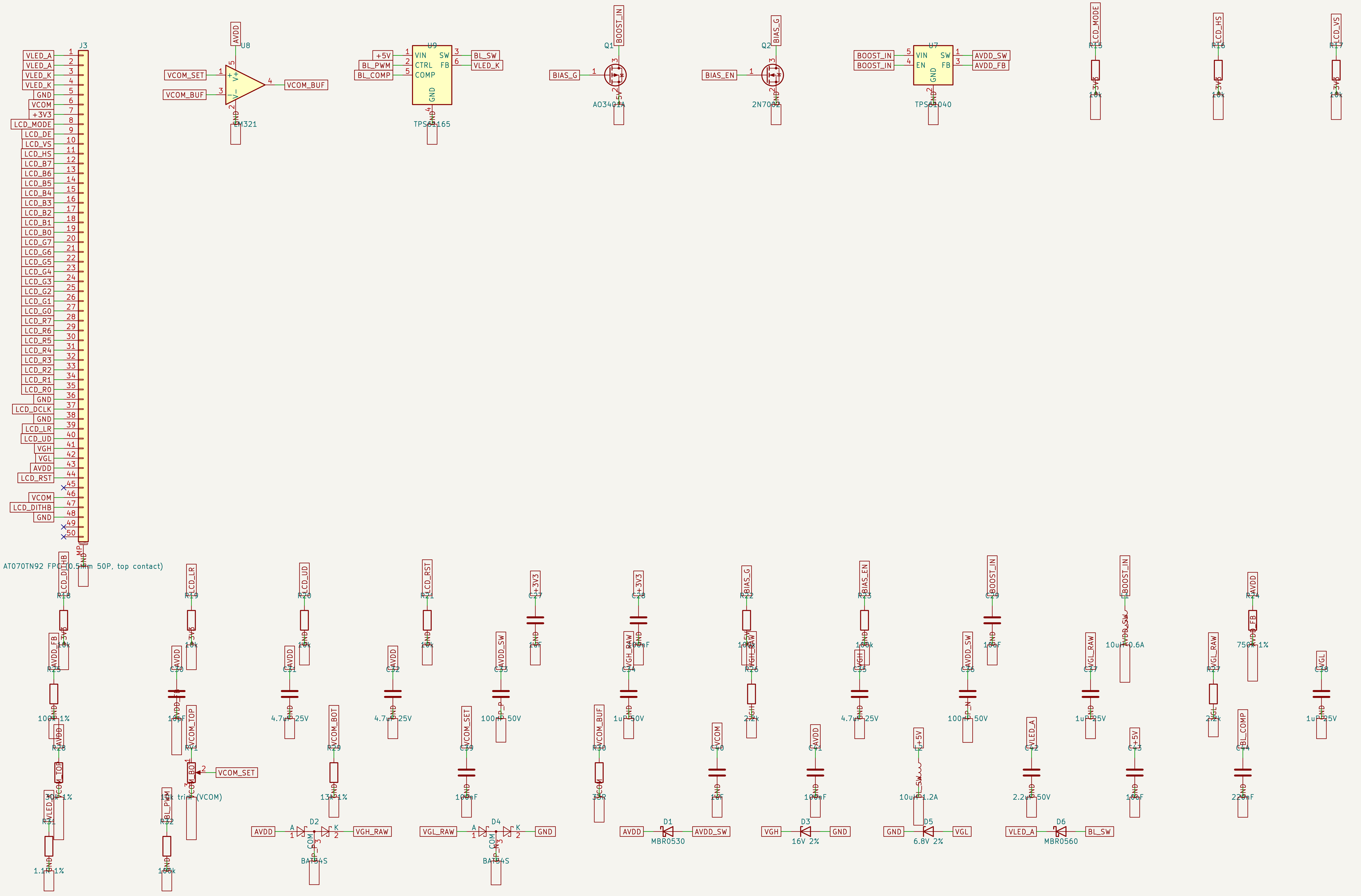
Generated from hardware/gen/design.py — edit the generator, not this file

AT070TN92 HDMI + USB touch driver

Sheet: /HDMI receiver LT8619C/
File: lcd_dual_hdmi.kicad_sch

Title: HDMI receiver LT8619C

Size: A3 Date: 2026-10-02 **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 3/5



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AT070TN92 HDMI + USB touch driver

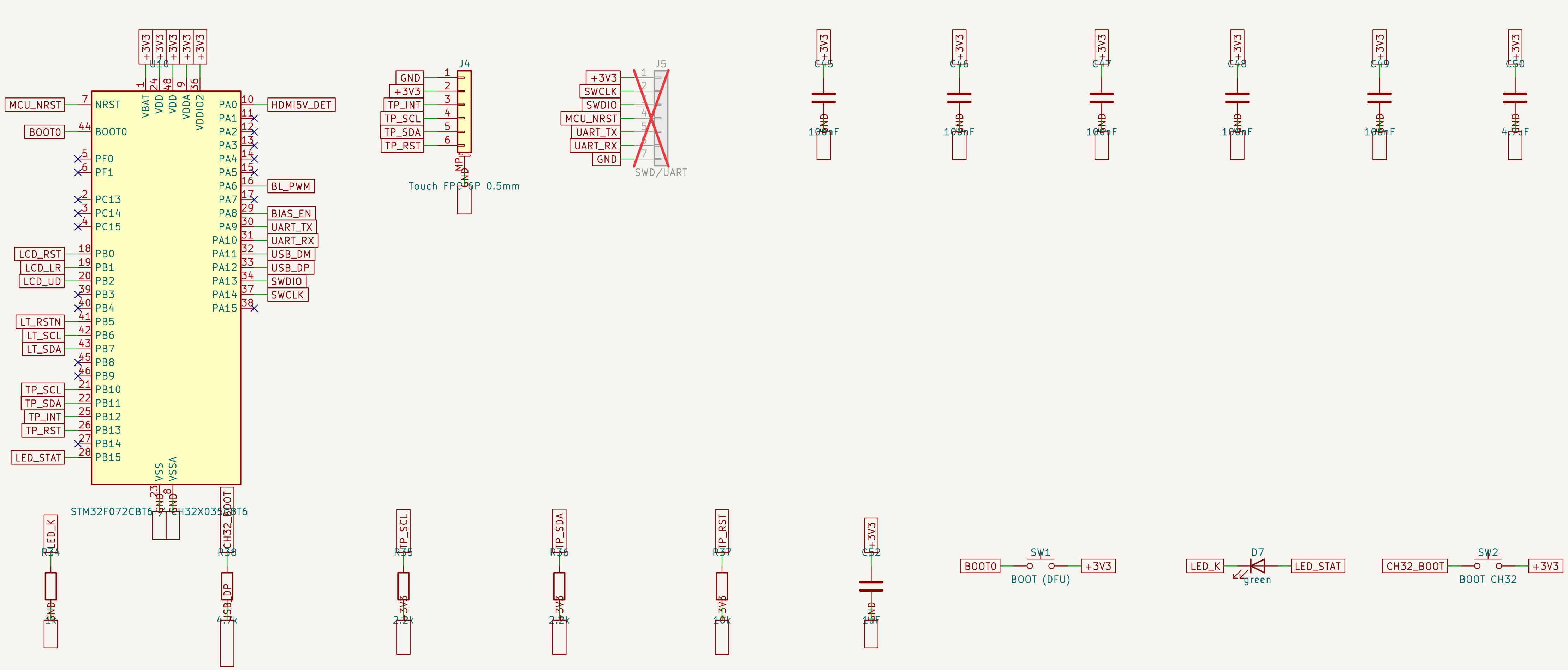
Sheet: /AT070TN92 panel, bias supplies, VCOM, backlight/

File: lcd_dual_panel.kicad_sch

Title: AT070TN92 panel, bias supplies, VCOM, backlight

Size: A3 Date: 2026-10-02 **Rev: 1.0**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/5



Generated from hardware/gen/design.py — edit the generator, not this file

AT070TN92 HDMI + USB touch driver

Sheet: /STM32F072 controller and GT915 touch/

File: lcd_dual_mcu.kicad_sch

Title: STM32F072 controller and GT915 touch

Size: A3

Date: 2026-10-02

Rev: 1.0

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 5/5